

Internship Report

AWS in a Smart Agriculture System

Introduction

In today's internship session, I learned about **Amazon Web Services (AWS)** and how cloud computing can be used in agriculture. I selected a **Smart Agriculture System** because it helps farmers monitor their farms, save water, and improve crop production using modern technology. AWS provides cloud services that make the system secure, reliable, and easy to manage. Farmers can monitor their fields from anywhere using a mobile phone or computer.

Why Cloud Computing?

Traditional farming requires farmers to visit their fields regularly to check soil moisture, weather conditions, and crop health. This takes time and may lead to delays in taking action.

By using AWS cloud services, farmers can:

- Monitor their farms remotely.
- Store farming data securely.
- Receive real-time notifications.
- Automate irrigation systems.
- Reduce water and electricity usage.
- Pay only for the cloud resources they use.

Cloud computing makes farming more efficient, cost-effective, and reliable.

Amazon EC2

Amazon EC2 is used to host the **Smart Agriculture System** application. Farmers can access the application through a web browser or mobile device to monitor their farms.

Using the EC2-hosted application, farmers can:

- View live sensor data.
- Check soil moisture, temperature, and humidity.
- Monitor crop health.
- Control irrigation systems.
- View farming reports and weather updates.

If many users access the application at the same time, the EC2 instance can be upgraded easily to maintain good performance without interrupting the service.

AWS Lambda

AWS Lambda performs automatic background tasks based on sensor data without requiring manual intervention.

For example:

- Automatically turns the water pump **ON** when soil moisture is low.
- Turns the pump **OFF** after the required moisture level is reached.
- Sends SMS or email notifications to farmers.
- Generates daily farming reports.
- Alerts farmers about high temperatures or low water levels.

Since AWS Lambda runs only when an event occurs, it reduces operational costs and improves the efficiency of the Smart Agriculture System.

Amazon S3

Amazon S3 is used to store agricultural data and files securely. Farmers generate many digital files such as crop images, drone photos, soil analysis reports, weather reports, and farming documents. These files can be stored safely in Amazon S3. It provides high durability, secure storage, and easy access to authorized users whenever needed. Older files can also be moved to lower-cost storage classes to reduce storage costs.

Database Services

The Smart Agriculture System stores important information such as farmer details, crop information, sensor readings, irrigation history, and weather data in a database.

- **Amazon RDS** is used to store structured data like farmer accounts, crop details, and reports.

- **Amazon DynamoDB** is used to store real-time sensor data because it provides fast and reliable performance.

These managed database services reduce maintenance work, improve reliability, and ensure data is always available.

IAM and Security

AWS Identity and Access Management (**IAM**) provides secure access to the Smart Agriculture System. Different users have different permissions based on their roles.

- **Farmers** can monitor sensor data and control irrigation.
- **Farm Managers** can manage multiple farms and view reports.
- **Administrators** can manage users, devices, and the complete system.

IAM ensures that only authorized users can access the system, protecting important farming data from unauthorized access.

Load Balancer and Reliability

If many farmers use the Smart Agriculture application at the same time, an **AWS Load Balancer** distributes incoming requests across multiple Amazon EC2 instances. This helps maintain smooth performance and prevents the application from becoming slow or unavailable.

AWS monitoring and backup services also help detect system issues, reduce downtime, and ensure that farming data remains safe and accessible even during unexpected failures.

Conclusion

From today's internship session, I learned that AWS can be used to build a Smart Agriculture System that is secure, scalable, and efficient. Services such as Amazon EC2, AWS Lambda, Amazon S3, Amazon RDS, Amazon DynamoDB, AWS IAM, and Load Balancer work together to help farmers monitor crops, automate irrigation, store farming data securely, and receive real-time notifications. This practical example showed me how AWS cloud services can improve modern farming by increasing productivity, reducing manual work, and making agriculture smarter.